

Aarjav Mitkari

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EDUCATION

Carleton University

Ottawa, ON

B. Eng in Software Engineering (Co-op option)

Sept. 2022 – Present

- Third year standing, 11.10/12 (A)
 - * Influential Courses: Introduction to Imperative Programming, Data Management, Discrete Structures 1, Discrete Structures 2, Object-Oriented Software Development, Algorithms and Data Structure
- 2024-25, 2023-24 and 2022-23 Dean's Honour List, Award of Excellence for International Students, Entrance Scholarship

EXPERIENCE

Software Developer Co-op

Jan 2025 – Dec 2025

Ribbon Communications

Ottawa, ON

- Built an automated database config readback system for six microservices to normalized XML format using Java (Spring Boot), Python (Flask), psycopg2, and PostgreSQL to allow easy comparison of snapshots across time and environments.
- Built declarative provisioning modules for the RAMP application, simplifying configuration management by 30%.
- Automated RAMP instance deployments on OpenShift by scripting Helm-based provisioning and configuration retrieval from Artifactory, cutting manual integration effort by 70%.
- Strengthened RAMP cloud platform security by contributing to pen-testing initiatives and identifying vulnerabilities and implementing fixes that reinforced platform's cloud defense and compliance standards.

AI/ML Research Assistant

May 2024 – December 2024

Carleton University

Ottawa, ON

- Collaborated with a professor on advanced AI/ML research, specializing in object tracking and computer vision.
- Developed algorithms with CNNs, transfer learning, data augmentation, and GANs for enhanced object detection and tracking.
- Improved model detection accuracy by 18% through hyperparameter tuning and dataset augmentation.

APPLIED PROJECTS

Digital Wallet Web Application | *Node.js, React.js, HTML5, CSS3, MongoDB*

June 2024 – July 2024

- Developed a full-stack digital wallet web application featuring user authentication with Zod, JSON Web Token (JWT) and Recoil for state management to implement real-time transaction processing.
- Designed dashboard for searching and transferring funds between user accounts with instant balance updates.
- Implemented robust backend logic to ensure secure and accurate financial transactions between user accounts.

Text Based Social Platform | *C*

November 2023 – December 2023

- Designed a command-line application in C to enhance the efficiency of user registration and management which optimized the storage and retrieval of user data by implementing ordered linked lists.
- Implemented an interactive CLI platform with dynamic linked lists to manage users and connections.
- Enabled $O(\log n)$ user lookup and duplicate prevention logic to maintain data integrity.

Stock Prediction Using Machine Learning | *Python, Pandas, Numpy, scikit-learn, XGBoost*

July 2024

- Built a predictive model using historical S&P 500 data from Yahoo Finance, leveraging both Random Forest and XGBoost classifiers to forecast stock movements.
- Implemented rolling averages & trend features, improving accuracy by 12% across test datasets.
- Conducted backtesting and tuning on 500K+ records, enhancing long-term prediction stability.

TECHNICAL SKILLS

Languages: Java, C, C++, JavaScript, Python, HTML/CSS

Framework & Concepts: Spring, Spring Boot, React, Node.js, Flask, Django, TailwindCSS, REST APIs, Microservices

Cloud & Infrastructure: Docker, Kubernetes, OpenStack, OpenShift, Helm, Jenkins, Terraform

Network Virtualization: CNF (Cloud-Native Network Functions), VNF (Virtual Network Functions) Architecture

Libraries: pandas, NumPy, Matplotlib, scikit-learn, PyTorch, TensorFlow, Java Swing, XGBoost

Agile Tools: JIRA, Git